

7.1.7 Software Qualification Testing Process

NOTE — The Software Qualification Testing Process in this International Standard is a lower-level process of the Software Implementation Process.

Users of ISO/IEC 15288 may decide that this process is to be provided by the Verification Process of ISO/IEC 15288 in a recursive application of that standard.

7.1.7.1 Purpose

The purpose of the Software Qualification Testing Process is to **confirm that the integrated software product meets its defined requirements** through formal testing conducted in a controlled environment.

7.1.7.2 Outcomes

As a result of successful implementation of the Software Qualification Testing Process:

- a) criteria for qualification testing of the integrated software are developed that demonstrate compliance with the software requirements;
- b) integrated software is verified using the defined qualification criteria;
- c) test results are recorded and stored as part of the project's audit trail; and
- d) a regression strategy is developed and applied for re-testing the integrated software when changes to software items occur.

NOTE — The regression strategy ensures that modifications to software items, requirements, design, or code trigger appropriate re-verification to maintain qualification status.

7.1.7.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Qualification Testing Process.

7.1.7.3.1 Software Qualification Testing

For each software item (or configuration item, if identified), the software

qualification testing activity consists of the following tasks:

7.1.7.3.1.0 Qualification Test Planning

The implementer shall define and document the **qualification test plan**, including:

- scope and objectives of qualification testing;
- mapping of qualification test cases to software requirements;
- responsibilities and schedule;
- test environments, tools, and data;
- acceptance criteria; and
- regression testing strategy.

The qualification test plan shall be reviewed, approved, and placed under configuration management.

7.1.7.3.1.1 Qualification Test Execution

The implementer shall conduct qualification testing in accordance with the qualification requirements and plan for the software item.

It shall be ensured that **the implementation of each software requirement is tested for compliance**. All qualification test results shall be documented and stored in /docs/audit/qualification/ or an equivalent controlled repository.

7.1.7.3.1.2 Documentation Update

The implementer shall update the user documentation as necessary to reflect the qualified software product.

7.1.7.3.1.3 Evaluation of Qualification Artifacts

The implementer shall evaluate the design, code, qualification test procedures, test results, and user documentation considering the following criteria. The results of the evaluations shall be documented:

- a) test coverage of all allocated software requirements;
- b) traceability between requirements, test cases, and results;
- c) conformance to expected results;
- d) completeness and accuracy of test procedures and data;
- e) adherence to applicable test standards and methods;
- f) readiness of the software item for subsequent system integration or acceptance

testing;
g) feasibility of operation and maintenance.

7.1.7.3.1.4 Regression Testing

The implementer shall **apply the regression testing strategy** to re-verify the integrated software whenever changes occur to software items, requirements, design, or code. Regression test results shall be recorded and stored with qualification evidence.

7.1.7.3.1.5 Audit Support

The implementer shall support audit(s) in accordance with subclause 7.2.7 (Software Audit Process).

The results of the audits shall be documented.

If both hardware and software are under concurrent development or integration, software qualification audits may be postponed until System Qualification Testing.

7.1.7.3.1.6 Handover Preparation

Upon successful completion of qualification testing and any required audits, the implementer shall prepare and update deliverables to support downstream processes such as System Integration, System Qualification Testing, Software Installation, or Software Acceptance Support, as applicable.

NOTE — The Software Qualification Testing Process may also be used to support the Software Verification Process (subclause 7.2.4) or the Software Validation Process (subclause 7.2.5).

7.1.7.4 Deliverables

The following deliverables shall be produced during the Software Qualification Testing Process:

- Qualification Test Plan
- Qualification test procedures, data, and environment specifications
- Qualification Test Reports and logs
- Regression testing strategy and results
- Updated user documentation
- Evaluation reports and traceability matrices

- Audit records (if applicable)
 - Qualification Audit Log (QAL)
 - Handover package for downstream system testing or acceptance activities
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7.1.7.5 Audit and Configuration Control

All qualification testing artifacts shall be placed under configuration management. Audit evidence (plans, procedures, results, logs, QAL entries) shall be retained for at least one release cycle or per applicable regulatory requirements. Digital signatures or equivalent mechanisms shall be applied to preserve the integrity and authenticity of qualification evidence.